

VOLUME 0 — PROJECT FOUNDATION

CHAPTER 0.1

Vision Document

Egocentric Wide-Angle Video Data Collection Platform

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1. Problem Statement

Modern computer vision, embodied AI, and robotics models depend on large volumes of realistic, first-person (egocentric), wide-field-of-view video captured in real-world environments. Public datasets are limited in scale, diversity, and viewpoint consistency, and most existing data-collection tools were built for narrow use cases (social video, surveillance, or consumer recording) rather than structured, task-driven dataset production.

Organizations that need this kind of data today are forced to improvise: asking field workers to record with their personal camera apps, manually renaming and transferring files, and manually tracking which recordings correspond to which task. This produces four recurring failures:

- Inconsistent capture — different zoom levels, orientations, and frame rates make footage unusable for training without manual filtering.
- Data loss — long single-file recordings that fail, get lost, or become corrupted before they ever reach a server.
- No traceability — nobody can reliably answer "which recordings belong to which project, task, and collector, and are they complete?"
- No resilience — recording and upload are treated as one blocking step, so collectors cannot keep working while previous footage uploads, and any network interruption stalls the entire pipeline.

Human Archive needs a purpose-built mobile application that removes these failure points by standardizing capture, automating chunking and upload, and giving both collectors and operations staff full visibility into the state of every recording session — from the moment a task is assigned to the moment the footage is verified in cloud storage.

2. Product Vision

Vision statement: To become the standard field-data-collection tool for structured, egocentric, wide-angle video datasets — enabling any distributed workforce to capture consistent, high-quality footage against assigned tasks, and get that footage safely into cloud storage without manual effort or data loss, regardless of network conditions.

The product is a Flutter-based mobile application, built for Android first, with iOS supported as configuration allows, that lets a "Collector" (the field worker or data-collection operator):

1. Log in securely (credentials or enterprise SSO) and see the Projects and Tasks assigned to them.
2. Review task instructions, reference examples, and requirements before recording.
3. Pass a pre-recording device health checklist (permissions, storage, battery, network).
4. Record continuous, high-bitrate, wide-angle (0.5x–0.6x) rear-facing video for a task.

5. Stop recording, at which point the app automatically splits the footage into 10-minute chunks and persists them locally.
6. Have those chunks uploaded automatically (or manually, if configured) in the background — resumable, and safe across app restarts and network loss — directly to cloud storage (AWS S3).
7. Immediately begin a new recording without waiting for the previous session's upload to finish.
8. See, at a glance, the sync status of every chunk and session, and retry anything that failed.

In later phases, the same platform extends to an Admin Web Dashboard for creating projects/tasks, assigning collectors, monitoring the collection fleet in real time, and reviewing/QA-ing the resulting dataset — turning the mobile app into one half of a complete data-operations platform.

3. Business Goals

- Enable reliable, scalable, repeatable video dataset collection to support AI/ML training pipelines that require egocentric, wide-angle footage.
- Eliminate data loss and corruption during field collection by moving from single monolithic recordings to automatically chunked, independently-persisted, independently-uploaded segments.
- Give operations and business stakeholders real-time visibility into collector throughput, task completion, and dataset completeness — without needing to manually audit files.
- Build a platform (not a one-off app) that supports multiple clients, multiple concurrent projects per collector, and configurable capture parameters — so it can be reused across future data-collection contracts.
- Lay the technical and product foundation for a future Admin Dashboard, QA/rework workflow, and collector performance analytics (Volume 3.3 and beyond).
- Minimize collector friction and training time — the app should be simple enough that a new collector can be onboarded and productive within a single session.

4. Success Metrics

The following metrics will be used to judge whether the product is achieving its vision. Exact targets will be finalized once a pilot cohort of collectors is running (see Volume 11 — Project Management).

- Upload success rate — percentage of recorded chunks that reach cloud storage without manual intervention (target: near-total reliability, tracked per week).
- Data-loss incidents — number of recorded chunks lost or corrupted before reaching cloud storage (target: zero, tracked per collector and app version).

- Chunk upload latency — median and 95th-percentile time from "chunk finalized locally" to "chunk confirmed in S3" under normal network conditions.
- Recovery time after network loss — time from network restoration to upload resumption, without any manual action from the collector.
- Collector active recording time — total hours of usable footage recorded per collector per day/week, as a throughput indicator.
- Task completion rate — percentage of assigned tasks reaching "Session Marked Complete" state.
- Crash-free session rate — percentage of recording sessions that complete without an app crash or force-close.
- Time-to-first-recording for a new collector — from app install to first successfully uploaded chunk, as an onboarding-quality indicator.

5. Long-Term Roadmap

The roadmap below is intentionally phased to match the volume structure of this documentation. Each phase only begins once the corresponding documentation volumes are complete and approved.

Phase 1 — MVP (Volumes 0–6 documentation → initial build)

- Core Features from Section 3.1: secure login, project/task retrieval, wide-angle rear-facing recording, automatic 10-minute chunking on stop, local persistence, automatic/manual upload, background resumable upload to S3, per-chunk metadata, offline queueing, session lifecycle, pre-recording health checklist, upload status visibility.
- Android-first release; iOS build pipeline evaluated in parallel (see Volume 7 — Development Environment).

Phase 2 — Secondary Features

- Session notes/tagging, task instructions with reference media, push notifications for upload failures and new assignments, multi-project support per collector, offline browsing of cached task/project data, in-app support contact, manual pause/resume/retry controls.

Phase 3 — Admin Platform (Future Phase, Section 3.3)

- Web-based Admin Dashboard for creating Projects/Tasks and assigning Collectors.
- Live upload/collection monitoring across the entire collector fleet.
- Dataset QA/review workflow with rework flagging.
- Collector performance and throughput analytics.
- Remote configuration of chunk duration, resolution, and bitrate via feature flags.

Phase 4 — Future Capabilities (Section 3.4)

- Over-the-air remote configuration of capture parameters without app-store releases.
- AI-assisted in-app QA — blur, occlusion, and motion-blur detection prior to upload.
- Multi-language localization for a global collector workforce.
- Simultaneous multi-camera capture (front + rear).
- Wearable capture device integration.

End of Chapter 0.1 — Vision Document